Energy-Aware LR-FHSS Multi-Beam Reception Under Satellite Power Constraints: A Derivation-Driven System Model

Abstract

Long-range frequency-hopping spread spectrum (LR-FHSS) reception on satellites is jointly constrained by demodulator availability and onboard electrical power limits. This draft presents a unified communication-energy model that couples traffic-dependent demodulator utilization, mode-dependent receiver power, electrical power subsystem (EPS) balance, and battery state-of-charge (SoC) evolution. The formulation starts from established equations in energy-proportional computing, queueing-based utilization, EPS power balance, and battery charging/discharging dynamics, and then derives a tractable recurrence suitable for time-step simulation. The resulting model enables joint analysis of decoding capacity and energy feasibility, which is essential when multi-beam operation must adapt to visibility and battery conditions. In reporting, we use decoded headers (header-only) as the primary KPI and decoded headers including payload-decoded packets as a secondary KPI.

1. Introduction

Multi-beam LR-FHSS reception improves uplink scalability, but each additional active demodulator increases receiver load and power demand. In orbit, available generation varies with visibility and operating state, making communication performance and energy feasibility inseparable. Existing studies provide strong component models for utilization-power relations [2], [3], spacecraft EPS balance [4], [5], and battery SoC propagation [1], [6], yet these components are often treated independently.

This work integrates these strands into one control-oriented recurrence that can be executed at simulation time steps. The receiver selects an operating mode, allocates demodulators, computes utilization and consumption, applies EPS balance, and propagates SoC to the next step. The key objective is not electrochemical detail, but a first-order model that preserves physically meaningful couplings between link activity and energy state.

The main contributions of this draft are: 1. A unified LR-FHSS communication-energy model for demodulator-constrained multi-beam reception under satellite power limits. 2. A transparent derivation linking literature-based equations into one implementable recurrence. 3. A battery-aware mode-control policy (`sleep`, `idle`, `busy`) that feeds back into decoding capacity and future energy state.

2. Problem Formulation

At each discrete time step t, the simulator receives node load N_t, requested demodulators $D_{\text{req},t}$, satellite visibility $V_t \in \{0, 1\}$, and battery SoC B_t. The task is to compute: 1. operating mode $m_t \in \{\text{sleep}, \text{idle}, \text{busy}\}$, 2. allocated demodulators D_t, 3. receiver consumption power $P_{\text{cons},t}$, 4. battery net power $P_{\text{net},t}$, 5. updated SoC B_{t+1}.

The overall state transition is:

$$(N_t, D_{\text{req},t}, V_t, B_t) \to (m_t, D_t, P_{\text{cons},t}, P_{\text{net},t}, B_{t+1}).$$

3. Model Derivation

3.1 Receiver Consumption Model From Energy-Proportional Form

Energy-proportional systems are commonly approximated by an affine law in utilization [2], [3]:

$$P(u) = P_{\text{idle}} + (P_{\text{peak}} - P_{\text{idle}})\, u, \quad u \in [0, 1].$$

For LR-FHSS reception, we apply this principle per allocated demodulator and allow mode-conditioned coefficients:

$$P_{\text{demod}}(u \mid m_t) = a_{m_t} + b_{m_t} u_t.$$

The total receiver consumption becomes:

$$P_{\text{cons},t} = \begin{cases} P_0, & m_t = \text{sleep}, \\ P_0 + D_t\,(a_{m_t} + b_{m_t} u_t)\,, & m_t \in \{\text{idle}, \text{busy}\}, \end{cases}$$

where P_0 is baseline platform power.

3.2 Utilization From Demand-Capacity Ratio

Queueing models use demand-to-capacity ratio as a utilization proxy:

$$\rho = \frac{\lambda}{c\mu}.$$

Using per-step traffic demand T_t, demodulator count D_t, and per-demodulator step capacity k, we define:

$$u_t = \min\left(1, \frac{T_t}{kD_t}\right), \quad D_t > 0,$$

and set $u_t = 0$ when $D_t = 0$. This clipping captures saturation while preserving bounded power behavior.

3.3 EPS Balance Under Visibility-Dependent Generation

Spacecraft EPS analysis uses generation-minus-load balance [4], [5]:

$$S_t = P_{\text{gen},t} - P_{\text{cons},t}.$$

Generation is modeled as:

$$P_{\text{gen},t} = V_t P_{\text{solar}} \eta_{\text{pc}},$$

where P_{solar} is panel power in illuminated condition and η_{pc} is power-conversion efficiency. Therefore,

$$S_t = V_t P_{\text{solar}} \eta_{\text{pc}} - P_{\text{cons},t}.$$

3.4 Battery Charge/Discharge Branches With Charge Acceptance

The CC-CV charging literature indicates tapering near full charge [6]. We represent this through an SoC-dependent acceptance factor $\alpha(B_t) \in [0, 1]$, monotonically decreasing as $B_t \to 100\%$.

Charging and discharging powers are:

$$P_{\text{ch},t} = \min\left(\max(S_t, 0),\ P_{\text{ch,max}} \alpha(B_t)\right),$$

$$P_{\text{dis},t} = \max(-S_t, 0),$$

$$P_{\text{net},t} = P_{\text{ch},t} - P_{\text{dis},t}.$$

3.5 SoC Update by Energy Integration

Following standard SoC integration [1], battery energy evolves as:

$$E_{t+1} = E_t + \Delta E_t.$$

For a step duration Δt (hours), with charging efficiency η_{ch} and discharging efficiency η_{dis}:

$$\Delta E_t = \eta_{\text{ch}} \max(P_{\text{net},t}, 0)\Delta t - \frac{1}{\eta_{\text{dis}}} \max(-P_{\text{net},t}, 0)\Delta t.$$

Given capacity C_{Wh} and $B_t = 100 E_t / C_{\text{Wh}}$:

$$B_{t+1} = \text{clip}_{[0,100]}\left(B_t + 100 \frac{\Delta E_t}{C_{\text{Wh}}}\right).$$

3.6 Control Policy and Demodulator Allocation

The control layer is represented as:

$$m_t = \pi(N_t, D_{\text{req},t}, V_t, B_t),$$

with allocation rule:

$$D_t = \begin{cases} 0, & m_t = \text{sleep}, \\ D_{\text{req},t}, & m_t \in \{\text{idle}, \text{busy}\}. \end{cases}$$

This policy creates a closed loop in which current battery and visibility determine demodulator availability, which in turn affects utilization, power draw, and future battery state.

4. Final Coupled Recurrence

Combining all blocks, the per-step evolution is:

$$(N_t, D_{\text{req},t}, V_t, B_t) \to m_t \to D_t \to u_t \to P_{\text{cons},t} \to S_t \to P_{\text{net},t} \to B_{t+1}.$$

An equivalent compact representation is:

$$B_{t+1} = \Psi(B_t,\ h(V_t P_{\text{solar}} \eta_{\text{pc}} - f(m_t, D_t, u_t),\ B_t))\,,$$

where $m_t = \pi(N_t, D_{\text{req},t}, V_t, B_t)$ and $u_t = \min\left(1, \frac{T_t}{kD_t}\right)$ for $D_t > 0$.

5. Implementation Mapping (Notation to Code)

Symbol	Meaning	Code variable / function
N_t	Node load	`nodes` in `run_workflow(...)`
$D_{\text{req},t}$	Requested demodulators	`requested_demods`
D_t	Allocated demodulators	`allocated_demods = allocate_demodulators(...)`
m_t	Power mode	`p_mode = select_satellite_power_mode(...)`
V_t	Visibility	`visible = check_satellite_visibility(...)`
T_t	Traffic demand proxy	`tx_count = transmit_fragments(packet_count, visible)`
$H_{\text{hdr},t}$	Header-only decoded count	`decoded_headers`
$H_{\text{inc},t}$	Headers including payload-decoded count	`decoded_headers_including_payloads`
u_t	Utilization	`compute_demod_utilization(tx_count, allocated_demods)`
$P_{\text{cons},t}$	Consumption power	`power_consumption = compute_power_consumption(...)`

Symbol	Meaning	Code variable / function
$P_{\text{gen},t}$	Generated power	`available_generation_watts` in `compute_power_balance(...)`
$P_{\text{ch},t}$	Charging power	`charging_power_watts`
$P_{\text{dis},t}$	Discharging power	`discharging_power_watts`
$P_{\text{net},t}$	Net battery power	`net_power_watts`
B_t, B_{t+1}	Battery SoC	`battery_percent_scenario`, `updated_battery_percent`
Δt	Step duration	`SIM_STEP_SECONDS / 3600.0`
C_{Wh}	Battery capacity	`BATTERY_CAPACITY_WH`

6. Scope and Limitations

The model is intentionally first-order and aimed at system-level tradeoff analysis. It does not explicitly include temperature-dependent internal resistance, detailed electrochemical states, or orbital-angle-resolved irradiance. These effects can be incorporated by replacing the generation and battery submodels while preserving the high-level recurrence and control-policy interface.

7. References

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